R Code

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# LAB 8: TIME SERIES FORECASTING USING ARIMA
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## To analyze quarterly CPI data using ARIMA and forecast CPI values
# for the next five years with appropriate prediction intervals.

# STEP 1: Load

# install.packages("readxl")
# install.packages("forecast")

library(readxl)
library(forecast)

# STEP 2: Import the Excel dataset

data <- read_excel(
  "C:/Users/sunita.sharma/Downloads/US_CPI_Quarterly_Real_Data.xlsx",
  sheet = "CPI Data"
)

# STEP 3: Explore the data

head(data)
tail(data)
str(data)
summary(data)

# Check missing values
colSums(is.na(data))

# STEP 4: Visualize the original data

plot(
  data$Date, data$CPI,
  type = "l",
  main = "Quarterly Consumer Price Index",
  xlab = "Year",
  ylab = "CPI",
  lwd = 2
)

# STEP 5: Convert data into a quarterly time series

# Data starts in 1959 Q1
# Frequency = 4 for quarterly observations

cpi_ts <- ts(
  data$CPI,
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  start = c(1959, 1),
  frequency = 4
)

plot(
  cpi_ts,
  main = "Quarterly CPI Time Series",
  xlab = "Year",
  ylab = "CPI",
  col = "blue",
  lwd = 2
)

# STEP 6: Examine ACF and PACF

par(mfrow = c(1, 2))

acf(cpi_ts, main = "ACF of CPI")
pacf(cpi_ts, main = "PACF of CPI")

par(mfrow = c(1, 1))

# STEP 7: Divide data into training and testing sets
# 80% training data and 20% testing data

n <- length(cpi_ts)
train_size <- floor(0.80 * n)

train <- window(
  cpi_ts,
  end = time(cpi_ts)[train_size]
)

test <- window(
  cpi_ts,
  start = time(cpi_ts)[train_size + 1]
)

length(train)
length(test)

# STEP 8: Fit an ARIMA model using training data

model <- auto.arima(
  train,
  seasonal = TRUE
)

# Display model results

summary(model)

# Display model coefficients

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coef(model)

# Compare information criteria

AIC(model)
BIC(model)

# Display selected ARIMA orders

arimaorder(model)

# STEP 9: Forecast the testing period

fc_test <- forecast(
  model,
  h = length(test),
  level = c(80, 95)
)

print(fc_test)

# Plot forecast and actual testing observations

plot(
  fc_test,
  main = "ARIMA Forecast for Testing Period",
  xlab = "Year",
  ylab = "CPI"
)

lines(test, col = "red", lwd = 2)

legend(
  "topleft",
  legend = c("Forecast", "Actual test data"),
  col = c("blue", "red"),
  lty = 1,
  lwd = 2
)

# STEP 10: Evaluate forecasting accuracy

accuracy_results <- accuracy(fc_test, test)

print(accuracy_results)

# ME   = Mean Error
# RMSE = Root Mean Squared Error
# MAE = Mean Absolute Error
# MAPE = Mean Absolute Percentage Error

# STEP 11: Check residual diagnostics

checkresiduals(model)
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# Examine whether residuals behave approximately
# like white noise and whether autocorrelation remains.

# STEP 12: Fit the final model using all observations

final_model <- auto.arima(
  cpi_ts,
  seasonal = TRUE
)

summary(final_model)

# STEP 13: Forecast the next five years

# Quarterly data: 5 years x 4 quarters = 20 quarters

forecast_5years <- forecast(
  final_model,
  h = 20,
  level = c(80, 95)
)

# Display forecast results

print(forecast_5years)

# STEP 14: Plot the five-year forecast

plot(
  forecast_5years,
  main = "Five-Year Forecast of Quarterly CPI",
  xlab = "Year",
  ylab = "CPI",
  col = "blue",
  lwd = 2
)

# STEP 15: Create a forecast table

forecast_table <- data.frame(
  Quarter = paste0(
    floor(time(forecast_5years$mean)),
    " Q",
    cycle(forecast_5years$mean)
  ),
  Forecast = as.numeric(forecast_5years$mean),
  Lower_80 = as.numeric(forecast_5years$lower[, "80%"]),
  Upper_80 = as.numeric(forecast_5years$upper[, "80%"]),
  Lower_95 = as.numeric(forecast_5years$lower[, "95%"]),
  Upper_95 = as.numeric(forecast_5years$upper[, "95%"])
)

# Display the table
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print(forecast_table)

# STEP 16: Save forecasts to the Downloads folder

write.csv(
  forecast_table,
  "C:/Users/sunita.sharma/Downloads/CPI_Forecast_Next_5_Years.csv",
  row.names = FALSE
)

# STEP 17: Plot actual versus fitted values

plot(
  cpi_ts,
  main = "Actual versus Fitted CPI Values",
  xlab = "Year",
  ylab = "CPI",
  col = "black",
  lwd = 2
)

lines(
  fitted(final_model),
  col = "red",
  lwd = 2
)

legend(
  "topleft",
  legend = c("Actual CPI", "Fitted CPI"),
  col = c("black", "red"),
  lty = 1,
  lwd = 2
)

# STEP 18: Display final model accuracy

print(accuracy(final_model))


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